Wormhole Formation in Matter-Coupled Vacuum Transitions: Gravitational Collapse, Horizon Constraints, and the Limits of Cosmological Rebirth

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Abstract

Rebirth scenarios invoke a wormhole carrying matter from a collapsing region into a new expanding one. We ask whether a density-triggered vacuum transition can form one when the only matter is a canonical scalar field coupled to dark matter, whose false vacuum has the observed dark-energy density. In this classical model the null energy condition holds, so no ordinary traversable throat can form: no null ray has a minimum of the areal radius. Non-traversable bridges and quantum or baby-universe continuations are not excluded. For a thin wall the Israel junction conditions give a finite-tension critical exterior mass, equal to $(1+q)R_{\mathrm{dS}}$ up to corrections of order σ^2, below which every regular configuration collapses; the equality of horizon areas there is a geometric coincidence of the limit. A spherical Einstein–scalar code checked against exact solutions identified trapped surfaces in 65 of 72 runs at the base resolution and in 6 more after refinement; one stayed unresolved. A persistent expanding false-vacuum core appears only when the initial data already contain a core larger than its own de Sitter radius; none develops in the 42 runs that start smaller, so the simulations show persistence, not creation.

At the observed dark-energy scale that core has a radius of 5.3 Gpc and a critical mass of 1.1 × 10^{23} solar masses, dominated by the vacuum-energy difference: today's matter in such a sphere is 23% of it. Matter is not transferred; a reset is not supported. The numerical evidence is complete only for anti-de Sitter exteriors.

Keywords

Wormholes; Spacetime throats; Vacuum decay; False-vacuum bubbles; Null energy condition; Gravitational collapse

1. Introduction

In a universe that dilutes its matter and approaches a cold, dark-energy-dominated state, any mechanism that returns the cosmos to a hot, dense phase has to do two things: change the vacuum, and supply a geometry through which a new expanding region can be reached. Cyclic models [1] and cosmological natural selection [2] place that geometry in a bounce or in a black-hole interior. A more literal proposal is a wormhole: a spacetime throat that forms near the end of the universe and carries matter back to a new beginning. General relativity does not establish that a wormhole behaves as a gravitational vacuum cleaner, and ordinary traversable wormholes need stress-energy that violates the null energy condition [3–5]. A defensible version of the idea must therefore be stated as a set of mathematical questions that can fail.

This paper states and answers such questions for one model. The model is a scalar field φ with a double-well potential whose false vacuum is held in place by dark matter through a symmetron-type coupling [6], so that the observed dark-energy density is carried by the false vacuum while the matter density is above a critical value, and released once the expansion has diluted the matter (Section 2). The release can start locally, by the nucleation of a bubble of the lower vacuum, and the reverse process, the return of a region to the false vacuum when its matter

density rises above the critical value, can also be local. The question is what the gravitational collapse of such a region can produce. The hypothesis, stated so that it can fail, is that a density-triggered vacuum transition may create a localized gravitational configuration that, under explicit conditions, develops a spacetime throat connecting a contracting region to a new expanding daughter region.

Three outcomes are distinguished throughout and are not allowed to blur. In outcome A the collapse is ordinary: a trapped region forms and no regular expanding region does. In outcome B a bridge-like geometry exists but nothing can cross it. In outcome C an expanding region exists behind a horizon. A collapsing region can produce a trapped surface without producing either a traversable wormhole or a new universe, and the distinction is the content of the paper. A second distinction is made as sharply: an expanding region that is present in the initial data and persists beside a collapse is not a daughter universe that the collapse has created. Even a created daughter would establish cosmological reproduction, not a cyclic return of the same universe, which additionally needs a daughter that reconnects with or replaces its parent; no part of this work provides that.

The mechanism is not new in its geometry. Thin false-vacuum bubbles in general relativity were analysed in detail in the 1980s [7,8], the possibility of a baby universe attached to a black hole by a throat was explored in classical and quantum form [9–11], and the obstacle that regular initial data do not lead to one was identified at the time [12]. What is new here is the application to a specific density-triggered model with a fixed vacuum scale. The vacuum energy is the observed dark-energy density, so the critical sizes and masses are numbers rather than free parameters, and they can be compared with the matter that the universe actually contains. The paper makes four contributions.

• A null-energy argument for the whole model (Section 3): no traversable throat can form, and no null ray has a minimum of the areal radius.

• A finite-tension thin-wall criterion (Section 4): a critical exterior mass $m_c(\sigma)$ below which a false-vacuum region is confined and collapses, with its small-tension limit $(1+q)R_{dS}$ and a coincidence of horizon areas at that limit (Section 6).

• A numerical Einstein–scalar experiment with a regular centre (Section 5), validated against exact solutions and by the convergence of its constraints, that tests the criterion with thick walls, separates expanding regions that persist from those that could be created, and compares exteriors of different vacuum depth.

• An accounting of the size, mass and energy budget that the criterion requires at the observed dark-energy scale, showing that the critical mass is dominated by the vacuum-energy difference and not by matter (Section 7), and falsification criteria (Section 8).

2. Model, inputs and scales

Units are $8\pi G = 1$ (M_{Pl} is the reduced Planck mass) and $c = \hbar = 1$ unless stated. The action is

$$S = \int d^4x\sqrt{-g}\left[\frac{M_{Pl}^2}{2}R - \frac{1}{2}(\nabla\phi)^2 - V_0(\phi)\right] + S_{dm}\left[A^2(\phi)g_{\mu\nu}, \psi_{dm}\right] \quad (1)$$

with $A(\varphi)=1+\varphi^2/2M^2$ the coupling to dark matter alone. For pressureless dark matter of conserved density $\rho_*\propto a^{-3}$ the scalar moves in $V_{eff}=V_0(\varphi)+\rho_* A(\varphi)$. While ρ_* exceeds a critical value $\rho_{crit}=\mu^2M^2$ the symmetric point of $V_0=V_c-\mu^2\varphi^2/2+\lambda\varphi^4/4$ is a minimum and holds the field in a false vacuum of density V_f. When the expansion has diluted the matter below ρ_{crit} the point becomes a maximum and the field can roll to a lower vacuum. Two numbers fix the gravitational problem, and nothing else of the particle physics enters it: the false-vacuum density V_f, which is set equal to the observed dark-energy density, and the depth of the true vacuum, $V_t=-qV_f$. The baseline is

q=1; q>0 is an anti-de Sitter true vacuum, q=0 a Minkowski one and −1<q<0 a lower de Sitter vacuum. The relevant length is the de Sitter radius of the false vacuum,

$$R_{dS} = \sqrt{3/V_f} = H_\Lambda^{-1} \tag{2}$$

which for the Planck 2018 values [13] H_0=67.66 km s^{-1} Mpc^{-1} and Ω_Λ=0.6889 is 5.34 Gpc (H_Λ=56.2 km s^{-1} Mpc^{-1}). The gravitational effect of a wall between the vacua is set by its tension σ. A scalar whose potential scale is the dark-energy scale, $V_f^{1/4}$≈2.2 meV, has a wall tension of the order of $V_f^{3/4}$, that is of order 10 meV^3; a bounce calculation for a potential of this kind gives 19 meV^3 and a thickness of 0.24 mm, and the conclusions depend only on the wall being a probe, not on these values. In units of the de Sitter radius this is $\hat{\sigma}=\sigma R_{dS}/M_{Pl}^2 = 2.7 \times 10^{-30}$ and $\delta/R_{dS} = 1.5 \times 10^{-30}$, so that the wall's own gravity, measured by $3\hat{\sigma}^2/4$ for q=1, is 5.5×10^{-60}. The wall is a probe, and its thickness is invisible to any gravitational observable of the size of the de Sitter radius. That fact organizes the paper: the thin-wall treatment of Section 4 is controlled by these two numbers, and the numerical experiments of Section 5 can only reach thick, heavy walls and therefore test the structure of the solution, not the physical values.

2.1 Inputs and what is derived here

The inputs taken from the benchmark dilution-triggered model are the false-vacuum density V_f (the observed dark energy), the depth q of the true vacuum (baseline 1, varied from −0.5 to 2 in Section 5.5) and the order of magnitude of the wall tension. Everything else is derived or computed in this paper: the energy-condition argument (Section 3), the junction analysis (Section 4), the numerical solver and its experiments (Section 5), and the size, mass and entropy accounting (Sections 6 and 7); none of it relies on a result of the benchmark model's cosmology. One limit of scope follows. The dark matter is not simulated: the numerical experiments evolve the scalar field alone in a prescribed double-well potential, so they study the gravitational

problem of a false-vacuum region bounded by a wall and not the complete matter-triggered process. The dark-matter coupling enters only through the existence of the false vacuum, and the claim that the whole process has been simulated is not made.

2.2 The exterior vacuum

The thin-wall model of Section 4 and the experiments of Section 5 place a false-vacuum region (positive density V_f) inside an exterior of lower vacuum energy. That exterior is not the late-time de Sitter state of the parent universe: the parent's heat-death state is the false vacuum itself. The exterior is the region in which the field has rolled to the true vacuum, which for $q>0$ is anti-de Sitter and recollapsing and for $q\leq 0$ is not. A false-vacuum region inside it arises when a region returns to the false vacuum, as when its matter density rises above ρ_{crit} and pins the field, while its surroundings have already made the transition; the nucleation of a bubble of the true vacuum inside the false vacuum [14] is the opposite configuration, whose interior is anti-de Sitter and ends in a crunch [15], so that it settles outcome A for that direction without further work. How such a region is reached, and whether a negative exterior is cosmologically viable, are not decided by this paper. What it can do is test how much the conclusions depend on the sign of the exterior vacuum energy: the thin-wall criterion holds for any $q>-1$ (Section 4), and the numerical experiments are repeated for a Minkowski exterior and a de Sitter exterior of lower energy in Section 5.5.

Table 1 lists the scales.

Quantity	Symbol	Value
de Sitter radius of the false vacuum	$R_{dS}=H_\Lambda^{-1}$	5.34 Gpc (1.65×10^{26} m)
Hubble rate of the false vacuum	H_Λ	56.2 km s^{-1} Mpc^{-1}
Wall tension in units of M_{Pl}^2/R_{dS}	$\hat{\sigma}$	2.7×10^{-30}
Wall thickness over R_{dS}	δ/R_{dS}	1.5×10^{-30}
Gravitational parameter of the wall	$3\hat{\sigma}^2/4$	5.5×10^{-60}
de Sitter entropy of the false vacuum	$\pi R_{dS}^2/\ell_P^2$	3.3×10^{122}
Matter within one Hubble radius today	M_H	$1.4 \times 10^{22}\ M_{odot}$

Table 1. Scales of the benchmark model (q=1, Planck 2018 cosmology).

3. What the energy conditions allow

A spherically symmetric spacetime is written $ds^2=-f(u,v)\,du\,dv+r(u,v)^2 d\Omega^2$ with null coordinates u (outgoing rays are $u=$ const) and v (ingoing rays are $v=$ const) and areal radius r. The null expansions of the two families are proportional to r_u and r_v. A sphere is normal if $r_v>0>r_u$, trapped if $r_v<0$ and $r_u<0$, and anti-trapped (expanding) if $r_v>0$ and $r_u>0$; the fourth quadrant, $r_u>0>r_v$, is the second exterior region of a bridge (a Kruskal-type extension). The Misner–Sharp mass [16] is

$$m = r\left(1+\frac{4\,r_u\,r_v}{f}\right) \tag{3}$$

in units where it includes the vacuum energy, $m=\rho r^3/3$ for a vacuum of density ρ, and a Schwarzschild–(anti-)de Sitter exterior of mass parameter $m_{\text{ext}}=2GM$ has $m=m_{\text{ext}}+\rho_{\text{out}} r^3/3$. Einstein's equations for the model reduce to (derived symbolically in the repository)

$$r_{vv}-\left(\ln f\right)_v r_v = -\frac{r}{2}T^{\text{dust}}_{vv}-\frac{r}{2}\left(\phi_v\right)^2 \tag{4}$$

$$r_{uv} = -\frac{r_u\,r_v}{r}-\frac{f}{4\,r}+\frac{U f r}{4} \tag{5}$$

where U is the total potential (the dust term is omitted in the numerical experiments, Section 5, where the field alone is evolved) and Eq. (4) is the constraint along v (and the analogue along u). The sign of the right-hand side of Eq. (4) is the whole story about throats. For every term of the model the null energy condition holds. For a null vector k, $T_{\mu\nu}k^\mu k^\nu=(k^\mu\partial_\mu\varphi)^2+\rho_{\text{dm}}(k\cdot u)^2\geq 0$: the potential terms $-g_{\mu\nu}V$ drop out because k is null, whatever the sign of V (including a negative true-vacuum density), and the dark matter is dust of non-negative density in the Einstein frame.

Proposition 1. In spherical symmetry with $T_{\mu\nu}k^\mu k^\nu\geq 0$, no null geodesic has a strict local minimum of the areal radius. Proof. A minimum along an outgoing ray requires $r_v=0$ and $r_{vv}\geq 0$ there. Eq. (4) at $r_v=0$ gives $r_{vv}=-(r/2)T_{vv}\leq 0$, so $r_{vv}\geq 0$ forces $T_{vv}=0$ and the minimum is not strict;

the same argument with u and v exchanged applies to ingoing rays. □ This is the local content of the focusing theorem. It means that a throat in the sense of Morris and Thorne, a minimal sphere that a traveller can cross, or a timelike marginal surface of Hayward's dynamic definition, needs matter that violates the null energy condition [4,5,17]. The model has none, so an ordinary traversable throat cannot form in it for any parameter values or any wall profile. The statement is limited to this classical model of a canonical scalar and non-negative-density dust: it does not exclude non-traversable bridges, quantum or semiclassical continuations that violate the null energy condition locally, or baby-universe scenarios that rest on them. Outcome B survives only as a non-traversable bridge: a minimal sphere on a spacelike slice (as in the Einstein–Rosen bridge, where the null energy condition holds trivially) that is hidden behind horizons, and outcome C only as an expanding region attached to such a bridge.

That settles the central obstacle identified at the outset, and it narrows the question to what such a bridge needs. Because the question does not involve exotic matter, it can be put to the ordinary Einstein–scalar system within a thin-wall treatment and numerically with a thick wall.

4. Thin-wall criterion

A thin shell of tension σ at areal radius $R(\tau)$, with $U=dR/d\tau$, separates an interior de Sitter region of density $\rho_{\rm in}=V_{\rm f}$ from an exterior Schwarzschild–(anti-)de Sitter region of mass parameter m and density $\rho_{\rm out}=-q\rho_{\rm in}$ [7,18]. With $\Delta=\rho_{\rm in}-\rho_{\rm out}=(1+q)V_{\rm f}$ the Israel junction condition takes the form

$$\Gamma_{\rm in}-\Gamma_{out}=\frac{\sigma R}{2},\qquad \Gamma_{\rm in}^2=1+U^2-\frac{R^2}{R_{dS}^2},\qquad \Gamma_{out}^2=1+U^2-\frac{m}{R}-\frac{\rho_{out}R^2}{3} \tag{6}$$

where $\Gamma=dR/dn$ is the derivative of the areal radius along the unit normal, which points from the interior to the exterior, so that a negative $\Gamma_{\rm in}$ means that the interior region extends beyond the equator of its de Sitter 3-sphere. Eliminating $\Gamma_{\rm out}$ gives

$$\Gamma_{\rm in} = \frac{\sigma R}{4} + \frac{m}{\sigma R^2} - \frac{\Delta R}{3\sigma} \tag{7}$$

Motion is possible only where $U^2 \geq 0$. For $R < R_{\rm dS}$ this reads $m \geq m_+(R)$ or $m \leq m_-(R)$ with

$$m_\pm(R) = \left(\frac{\Delta}{3} - \frac{\sigma^2}{4}\right) R^3 \pm \sigma R^2 \sqrt{1 - R^2/R_{dS}^2} \tag{8}$$

and for $R \geq R_{\rm dS}$ it is always satisfied. The curves $m_\pm$ bound a band of forbidden radii (Fig. 1). The band reaches up to the finite-tension critical mass

$$m_c(\hat{\sigma}) = \max_{R \leq R_{dS}} m_+(R) = (1+q) R_{dS} - \hat{\sigma}^2 \left[\frac{1}{4} - \frac{1}{6(1+q)}\right] R_{dS} + O(\hat{\sigma}^4) \tag{9}$$

which is exact within the junction equations (the maximum is found numerically; the expansion is that of the maximum of Eq. (8) near $R = R_{\rm dS}$ and is checked against it in Table 2). The simple form $m_{\rm c} = (1+q) R_{\rm dS}$ is the limit of vanishing tension, not an additional result: its relative error is of order $\hat{\sigma}^2$ (6.1×10^{-61} for $q = 1$ and the model's tension), and the two are indistinguishable there. For $m < m_{\rm c}$ every wall lies either at $R \leq R_1(m)$ or at $R \geq R_2(m)$, with a forbidden window $R_1 < R < R_2$ around $R_{\rm M} = (3m/\Delta)^{1/3}$ that no trajectory crosses. For $m > m_{\rm c}$ there is no window. A numerical scan of R confirms this at nine combinations of tension and depth: a window exists at $m = m_{\rm c}(1 - 10^{-3})$ and none at $m = m_{\rm c}(1 + 10^{-3})$ in every case (Table S0 of the supplementary information).

$\hat{\sigma}$	$m_{\rm c}(\hat{\sigma})$	Expansion, Eq. (9)	$r_{\rm h}(m_{\rm c})$	$S_{\rm dS}/S_{\rm BH}$
0.01	1.99998	1.99998	1.00000	1.00001
0.10	1.99833	1.99833	0.99958	1.00083
0.30	1.98505	1.98500	0.99625	1.00754
0.90	1.86927	1.86500	0.96648	1.07056
2.7×10^{-30} (model)	$2 - 1.2 \times 10^{-60}$	$2 - 1.2 \times 10^{-60}$	$1 - 3.1 \times 10^{-61}$	$1 + 6.1 \times 10^{-61}$

Table 2. Finite-tension critical mass for $q = 1$ (units of $R_{\rm dS}$): the numerically maximized $m_{\rm c}(\hat{\sigma})$, its small-tension expansion of Eq. (9), the radius of the exterior horizon at $m_{\rm c}$ and the ratio of the de Sitter and black-hole horizon entropies, $(R_{\rm dS}/r_{\rm h})^2$, which equals 1 only in the limit.

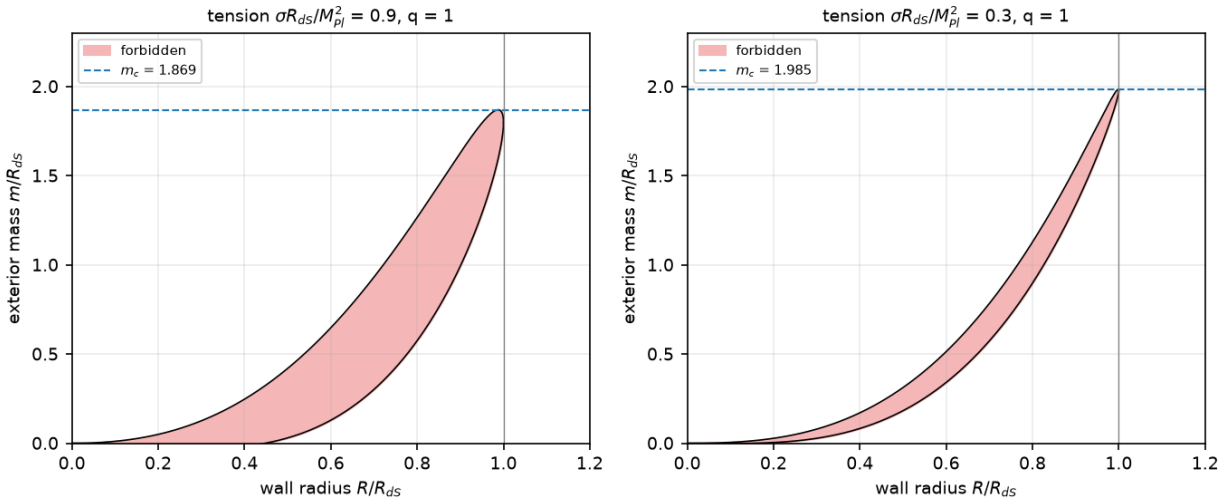


Fig. 1. Allowed and forbidden wall radii for q=1 and two values of the wall tension. The shaded band is forbidden; the dashed line is the finite-tension critical mass $m_c(\sigma)$. For $m<m_c$ a horizontal line crosses the band, which a wall cannot cross; for $m>m_c$ it does not.

The consequences follow from which side of the window a wall can be on and what a regular initial slice looks like. A slice that is regular, in the sense that the interior is a de Sitter ball less than half of its 3-sphere ($\Gamma_{in}>0$) and the exterior is an asymptotic region ($\Gamma_{out}>0$), has $R<R_M(1-O(\sigma))$. If $m<m_c$ it therefore lies at $R\leq R_1$, can never reach R_2, and collapses to $R=0$, where the singularity theorem applies [19]: outcome A for every regular configuration of subcritical mass, with no assumption on the initial velocity. The branch $R\geq R_2$ needs $\Gamma_{in}<0$, an interior beyond the equator of its own de Sitter space, which is not a regular ball; that is the configuration of the baby-universe solutions of Refs. [7,10], and it cannot be reached from regular data, which is the obstacle of Ref. [12]. For $m>m_c$ the window is absent, so a wall that starts regular can move out through R_M, where Γ_{in} changes sign.

The bridge in these solutions needs to be stated precisely, because the wall is not where the minimal sphere is. On the time-symmetric slice through a turning point with $\Gamma_{in}<0$ and $\Gamma_{out}<0$, the areal radius is zero at the pole of the de Sitter cap, rises to R_{dS} at its equator (a maximal sphere),

decreases across the wall, and continues to decrease through the exterior to its minimum at the bifurcation sphere of the exterior solution, $r=r_h$, beyond which it increases again to the second asymptotic region. The minimal sphere is therefore the bifurcation sphere of the exterior's horizons, which is a marginal surface and not a throat that can be crossed; the bridge is non-traversable, as Proposition 1 requires. This construction rests on the thin-wall solution and its literature: the numerical evolution of Section 5 starts on a null cone and ends at the singularity, and does not construct such a slice, so no throat is claimed for it.

Two statements follow. First, $m>m_c(\sigma)$ is necessary for any regular configuration to reach $\Gamma_{in}<0$. Second, because m_c is fixed by the vacuum energy through R_{dS}, the necessary mass is a property of the observed dark-energy scale, not of the particle physics. Whether the condition is also sufficient, and what thick walls do, is a question for the numerical evolution of the next section.

5. Numerical experiment: collapse of a false-vacuum core with a regular centre

5.1 Formulation

The thin-wall result assumes an infinitely thin, infinitely light wall. To test its structure with a wall of finite thickness and tension, the coupled Einstein–scalar equations are evolved in double-null gauge with a regular centre, the standard formulation of spherical scalar-field collapse [20,21], which resolves trapped surfaces and expanding regions and requires no choice of slicing near horizons. The equations are Eqs. (4) and (5) with the scalar equation and the equation for f,

$$\phi_{uv} = -\frac{\phi_u r_v + \phi_v r_u}{r} - \frac{f}{4} U'(\phi), \qquad (\ln f)_{uv} = \frac{f + 4 r_u r_v}{2 r^2} - \phi_u \phi_v \tag{10}$$

with $U(\varphi)=(\kappa/4)(\varphi^2/A^2-1)^2+s\varphi/A+d$, tuned so that the false minimum has density $\rho_{in}=3$ (so that $R_{dS}=1$) and the true minimum has density $-q\rho_{in}$, with $\kappa=24$ and, in the baseline, $q=1$. The field amplitude A is the only free scale of the wall: its thickness is $\ell=A\sqrt{2}/\kappa$ and its tension is of the order of $A\sqrt{\kappa}/2$, so that $A\to 0$ is the thin, light wall limit. Only the scalar field is evolved; the dark matter of the full model is not included (Section 2.1). The scheme is the second-order diamond predictor–corrector on a uniform lattice [22,23]: r and φ are updated cell by cell, the metric function $\sigma=\ln f$ is integrated from the constraint within six points of the centre, where the evolution equation divides by a small radius, and from its evolution equation beyond; the centre value of σ follows from elementary flatness, $r_v=\sqrt{f}/2$, solved by the secant method because the plain iteration has a gain of modulus greater than one, and the centre value of φ from regularity, $\varphi_t=2\varphi_v$, by an implicit trapezoid. Data are given on the outgoing null cone from the centre in the gauge $r=v/2$: $\varphi(0,v)$ is false vacuum for $v<V_w$ and true vacuum beyond, with a tanh transition of width w, and $\sigma(0,v)$ follows from the constraint. The wall is then an ingoing wall of radius $R_w=V_w/2$, with the false-vacuum core inside it and no matter outside. The code is dnull.py; the equations were derived symbolically (derive_equations.py).

Three limits of the method should be stated at once. The scheme does not follow the evolution beyond the formation of a singularity, where f diverges: a run ends when σ exceeds 12 or a value becomes non-finite, and the classification below uses only features that cannot be undone by the evolution that follows, a trapped surface and an anti-trapped region. The wall is far thicker and heavier than the physical wall by thirty and sixty orders of magnitude (Table 1), so the experiments test structure, not the values of the physical model. And the evolution is spherically symmetric and classical.

5.2 Verification

The solver was checked against the two exact vacuum solutions with a regular centre. For de Sitter space ($\rho=3$) and anti-de Sitter space ($\rho=-3$) the Misner–Sharp mass of Eq. (3) must equal $\rho r^3/3$ at every point. Table 3 gives the largest relative error for three grid steps; the errors fall at second order. The radius of the first trapped point of a collapsing run ($R_w=1$, $w=0.1$, $A=0.1$) converges at the same order, 1.0859, 1.0837 and 1.0831 at $h=0.02$, 0.01 and 0.005, with a Richardson extrapolation of 1.0829.

Vacuum	$h=0.04$	$h=0.02$	$h=0.01$	Orders
de Sitter	1.9×10^{-4}	5.1×10^{-5}	1.3×10^{-5}	1.9, 2.0
anti-de Sitter	1.8×10^{-4}	4.8×10^{-5}	4.8×10^{-6}	1.9, 3.3

Table 3. Verification against exact vacuum solutions: largest relative error of the Misner–Sharp mass with a regular centre ($u\leq0.6$, $v\leq3$).

Exact vacuum solutions do not test the dynamics of a scalar wall, so the constraints were evaluated on the dynamical solutions. The constraint of Eq. (4) along v and its analogue along u are imposed only within six points of the centre, and elsewhere the evolved fields are free to violate them. Their root-mean-square residuals over the region $0.05\leq u\leq0.28$, $u+0.4\leq v\leq3$ of the run with a core inside its de Sitter radius ($R_w=1$) fall with the grid step at orders 1.4 and 1.8 for C_{vv} and 2.3 and 2.1 for C_{uu} between $h=0.02$, 0.01 and 0.005, and for the run with a core outside it ($R_w=2$) at orders 1.7 and 1.9 for C_{vv} and 3.3 and 2.1 for C_{uu} (Fig. S1). The residuals are absolute (units $R_{dS}=1$); for scale each is also given relative to the RMS of the largest of the three terms of that constraint over the same region. For the core inside its de Sitter radius C_{vv} has RMS 5.0×10^{-1} (34% of its terms) at $h=0.02$, 1.8×10^{-1} (12% of its terms) at $h=0.01$ and 5.4×10^{-2} (3.3% of its terms) at $h=0.005$ and C_{uu} 1.2×10^{-2} (1.9% of its terms), 2.4×10^{-3} (0.36% of its terms) and 5.6×10^{-4} (0.08% of its terms); the largest values lie at the wall, whose width of 0.1 is resolved by about 5, 10 and 20 points. For the core outside it C_{vv} is 3.8×10^{-3} (2.7% of its terms), $1.2 \times$

10^{-3} (0.77% of its terms) and 3.2×10^{-4} (0.20% of its terms) and C_{uu} 1.3×10^{-3} (0.09% of its terms), 1.3×10^{-4} (0.01% of its terms) and 3.1×10^{-5} (0.00% of its terms). The base-resolution scans use h=min (0.01,w/8), so that the middle values apply to the thinner walls: at that resolution the constraint along v is satisfied near the wall to about 12% of its terms, and quantities that depend on the detail of the wall, such as the exact radius of a first trapped point inside a wall that is still passing, carry that uncertainty, whereas the exterior mass and the horizon structure are less sensitive (Section 5.3). The null expansions converge as well: the maximum difference of r, r_u and r_v at u=0.2 between successive resolutions falls with Cauchy ratios 4.1, 4.0 and 2.4 for the core inside its de Sitter radius (the ratio of r_v is lower, about order 1.3, because of the sharp wall) and 3.3, 3.0 and 3.2 for the core outside it, at u=0.12 and with the same signs of r_u and r_v at all points.

5.3 Horizon formation

A scan of 72 runs varies the field amplitude (A=0.05, 0.1 and 0.2), the wall width (w=0.05, 0.1 and 0.2) and the core radius R_w/R_{dS} from 0.4 to 3 (Supplementary Table S1). Each run measures the exterior mass m_{ext} on the initial cone beyond the wall, from the Misner–Sharp mass (constant across the exterior to within 2% in 70 of the 72 runs, at most 9.5%), and the radius of the first trapped point. At the base resolution a trapped surface forms before the code breaks down in 65 runs, and 7 runs ended without one. These seven were examined individually (Table S2): the code stopped on non-finite values with σ still between 1.5 and 2.8, far from the divergence that marks a singularity, so that the stops are numerical. Refinement by factors of 2 and 4 gave a trapped surface in 6 of them, so that trapped surfaces form in 71 of the 72 runs; the remaining run (A=0.1, w=0.1, R_w=0.4R_{dS}, a mass whose horizon radius, 0.2, is about two wall widths) stays unresolved at the finest step and is reported as such. The radius of the first trapped

point agrees with the horizon radius of the Schwarzschild–anti-de Sitter solution of the measured mass, $r_h^3+r_h=m_{ext}$ for q=1, to a median of 0.2%, a 95th percentile of 3.1% and a maximum of 6.2% over the 65 base-resolution runs (Fig. 2). The agreement is best at A=0.1 (median 0.1%) and is not monotonic in A, which is expected because the apparent horizon forms in a region where the wall has not yet fully passed; for the wide high-mass walls resolved only after refinement, the first trapped point lies 19% to 29% inside the final horizon radius for the same reason. This agreement checks the identification of the horizon against the expected exterior solution. It does not validate the interpretation of the expanding core as a separate region, which rests on the quadrant structure and on the persistence of the expanding core discussed next.

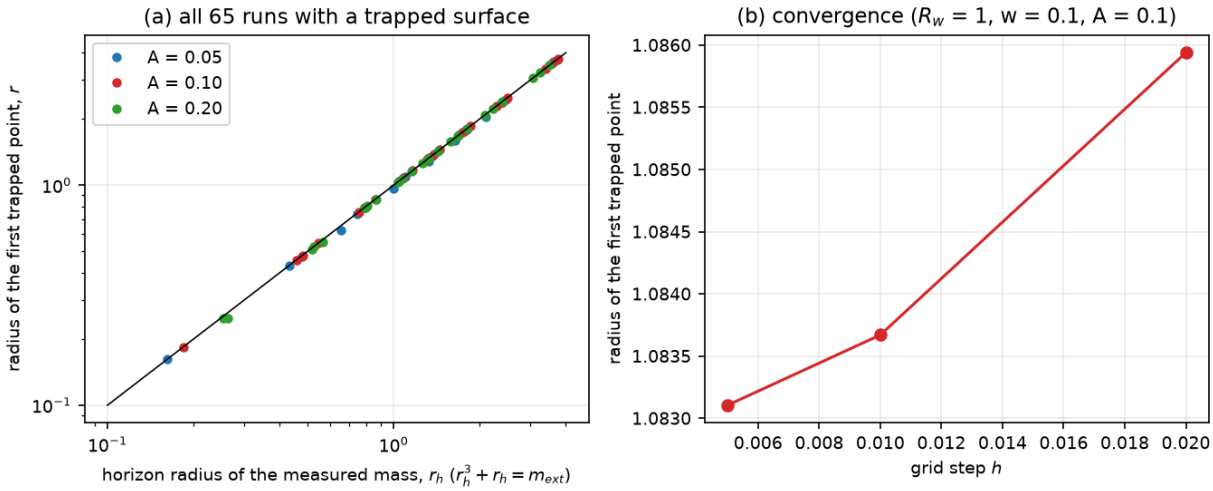


Fig. 2. (a) Radius of the first trapped point against the horizon radius of the Schwarzschild–anti-de Sitter solution of the mass measured outside the wall, for all runs with a trapped surface at the base resolution; the line is equality. (b) Convergence of the first trapped point with the grid step.

5.4 Expanding core: persistence or creation

The question that decides between outcomes A and C is whether an anti-trapped region survives next to the trapped surface, and the question that decides between persistence and

creation is whether it can appear where the initial data have none. Fig. 3 classifies the runs by core radius and exterior mass. A core that stays inside its own de Sitter radius ($R_w \leq 1.0\, R_{dS}$ for the thin walls, w=0.05, and $R_w \leq 1.2\, R_{dS}$ for the thicker ones, the offset being of the order of the wall width) is normal everywhere inside the wall in the initial data, and the collapse is outcome A with a trapped region and a shrinking regular core (37 runs with a trapped surface at the base resolution). A core that extends beyond its own de Sitter radius ($R_w \geq 1.2\, R_{dS}$ for thin walls, $\geq 1.5\, R_{dS}$ for thick walls) contains a region with $r_u > 0$ and $r_v > 0$ in the initial data, which persists to the end of every such run, next to a trapped region: 28 runs. The thin-wall criterion is a necessary condition in every run: all of the runs with an expanding core have $m_{ext} > m_c$. Mass alone does not decide the outcome, however: several runs with m_{ext} up to 3.0 times m_c and a core inside the de Sitter radius have no expanding region.

The distinction between persistence and creation can be tested directly, and it is the main limitation of the evidence. Of the 72 runs, 42 start with a core that is normal everywhere in the initial data and 30 start with an expanding core. In none of the 42 runs that start normal does an anti-trapped point appear at any later row. The simulations therefore show the persistence of an expanding false-vacuum core that the initial data already contain, beside a trapped region that forms around it; they do not show the dynamical creation of an expanding region, let alone a daughter universe, from initially non-expanding, localized data, and for such data the evidence is the opposite. Because the evolution stops before the global causal structure is established, ‘daughter universe’ is reserved in this paper for a hypothetical global interpretation of the persistent core.

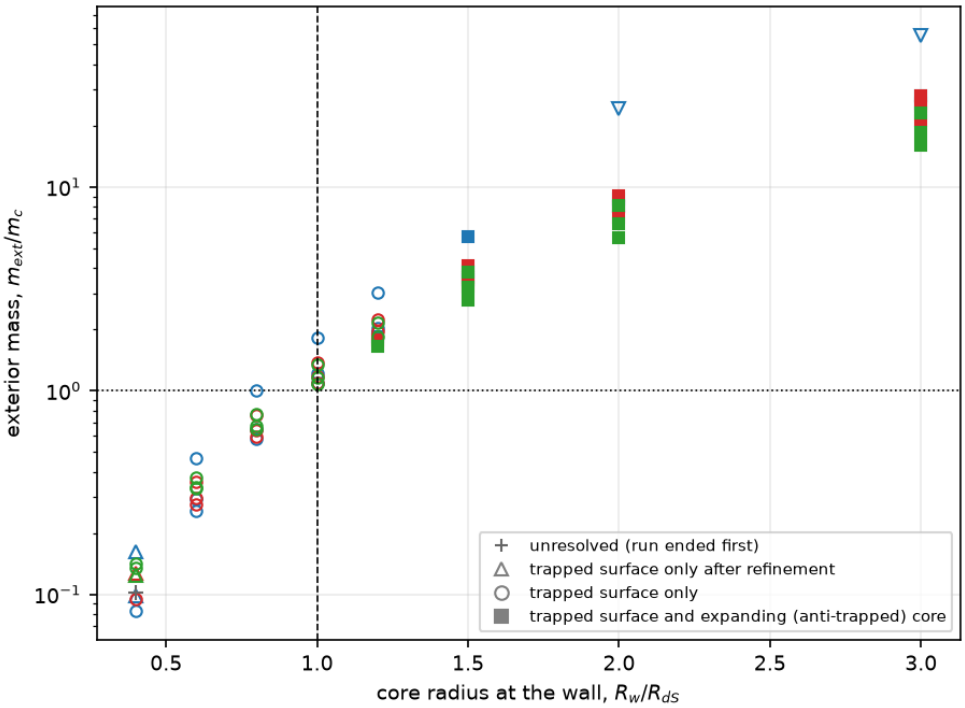


Fig. 3. Outcome of each run against core radius and exterior mass in units of the thin-wall critical mass. Open circles: a trapped surface forms and the core is normal (outcome A); filled squares: a trapped surface forms and an expanding core, present in the initial data, persists; open triangles: a trapped surface forms only after refinement; the cross: unresolved. The dashed line is the core radius equal to R_{dS} and the dotted line is $m=m_c$. Colours distinguish $A=0.05$ (blue), 0.1 (red) and 0.2 (green).

Fig. 4 shows the causal structure of two runs in terms of the signs of the null expansions. For $R_w=0.8R_{dS}$ (panel a) the region inside the ingoing wall stays normal and the region outside it becomes trapped behind the apparent horizon at $r\approx0.79$. For $R_w=2R_{dS}$ (panel b) a region of positive expansions in both directions fills the core between the de Sitter horizon and the wall; beyond the wall there is first a region with $r_u>0>r_v$, the second-exterior quadrant of a bridge, and then a trapped region. No null ray has a minimum of the areal radius, as Proposition 1 requires: over all 3297 rows of all 72 runs, no outgoing ray has a point at which r_v changes from negative

to positive. The runs do not construct a spacelike slice through the bridge, so no throat is claimed from them; the interpretation of the thin-wall bridge given in Section 4 rests on the exact solution.

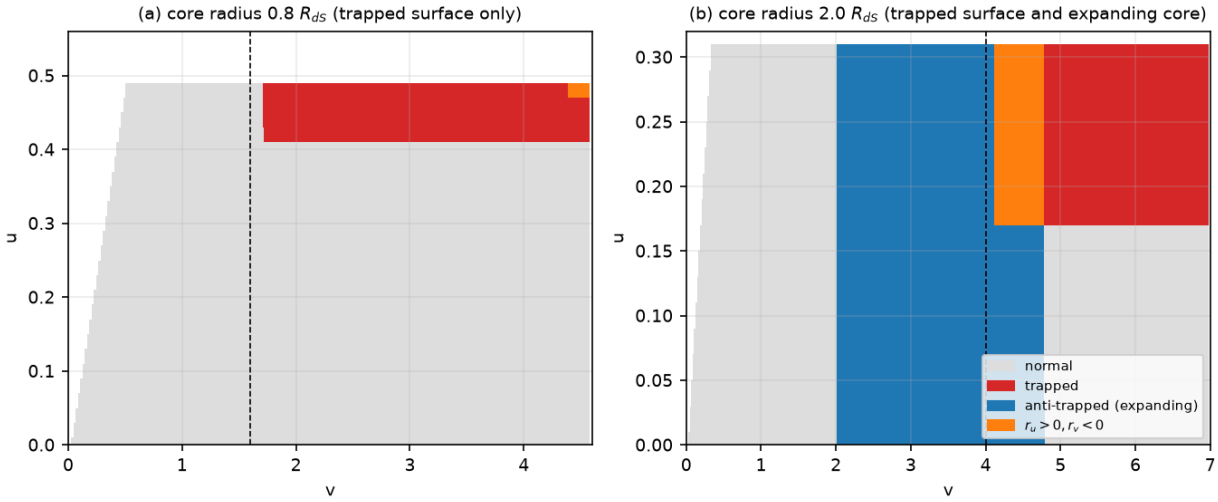


Fig. 4. Quadrants of the null expansions in the (u,v) plane for two runs (A=0.1, w=0.1): (a) a core inside its de Sitter radius, (b) a core outside it. The dashed line is the wall. Grey: normal ($r_u<0<r_v$); red: trapped; blue: anti-trapped (expanding); orange: $r_u>0>r_v$. The runs stop when the code can no longer follow the trapped region.

Two limits of this evidence should be explicit. First, it establishes that the structure predicted by the thin-wall theory is present at finite thickness and tension; it does not establish what lies beyond the horizon after the code ends, so the statement that an expanding core persists rests on the anti-trapped region, which the dynamics cannot undo, and not on a global evolution. Second, within the scan the thin-wall limit is approached but not reached; the thickness scale of the wall ranges from 1.4% to 5.8% of $R_{\rm dS}$ (Table 1 gives 10^{-30} for the physical wall), so that the match of the apparent-horizon radius to the analytic value to a fraction of a percent is a test of the code and of the exterior solution, not of the thin-wall limit itself.

5.5 Exteriors of other vacuum depth

The baseline exterior is anti-de Sitter (q=1). To test how much of the picture depends on that choice, the scan was repeated with A=0.1, two wall widths and seven core radii for q=2 (a deeper anti-de Sitter exterior), q=0 (a Minkowski exterior) and q=−0.5 (a de Sitter exterior of energy $0.5V_f$), with the thin-wall threshold $m_c=(1+q)R_{dS}$ in each case (Fig. 5, Supplementary Table S3). For q=2 the picture of the baseline is reproduced: a trapped surface forms in 14 of the 14 runs, with a radius within a median of 0.2% (maximum 1.3%) of the horizon radius of the exterior, and an expanding core is present in the 7 runs whose core extends beyond its de Sitter radius, none of them with a mass below m_c. For q=0 trapped surfaces were found in 5 of the 14 runs (the others ended on non-finite values before one formed), within a median of 0.5% of the Schwarzschild radius $r_h=m_{ext}$, and again an expanding core in the 7 runs with a core beyond R_{dS} and none below m_c. For q=−0.5 the exterior has a black-hole horizon only for masses below the Nariai value m=0.544 R_{dS}, and 10 of the 14 runs have a larger mass; no trapped surface was found in any of them, and an expanding core is present in the 7 runs with a core beyond R_{dS}, again with $m_{ext}>m_c$ in all.

Two conclusions follow, and a third limits them. The necessary condition $m>m_c$ and the dependence of the expanding core on a core larger than its de Sitter radius do not depend on the sign of the exterior vacuum energy in these runs. What does depend on it is the horizon structure: for $q\geq0$ the false-vacuum region is hidden behind a black-hole horizon, whereas for $q<0$ and large mass the exterior has no black-hole horizon at all, so that the configuration is not a daughter behind a horizon but an expanding region bounded by the exterior's cosmological horizon, and outcome C in the sense of this paper does not apply. The third is the limit: the q=0 and q=−0.5 scans are largely unresolved with respect to trapped surfaces, so that they support the

statement about the expanding core and the threshold, not a full classification. The distinction is the following. The thin-wall criterion of Section 4 holds for every $q>-1$ and so covers all of the vacuum depths modelled. The numerical evidence is strongest for the anti-de Sitter exteriors (q=1 and 2), where trapped surfaces are found in nearly all runs; the Minkowski and lower-energy de Sitter exteriors do not yet provide equally complete numerical confirmation.

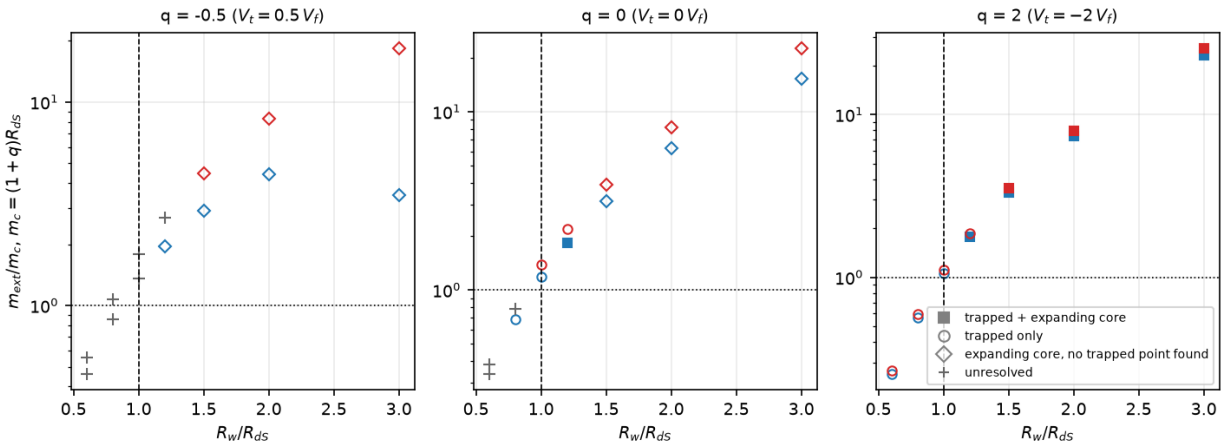


Fig. 5. Outcomes for exteriors of other vacuum depth (A=0.1; blue w=0.05, red w=0.1): core radius against exterior mass in units of $m_c=(1+q)R_{dS}$. The dashed line is $R_w=R_{dS}$ and the dotted line is $m=m_c$. Trapped surfaces are found in nearly all runs for q=2 (and for q=1, Fig. 3), in 5 of 14 for q=0 and in none of 14 for q=−0.5, where many masses exceed the Nariai value; 'unresolved' marks runs that ended before classification. The thin-wall criterion covers all three exteriors; the numerical confirmation is complete only for the anti-de Sitter case.

6. Matter transfer and a horizon-area coincidence

The plan to carry all of a universe's matter through a throat to a new beginning needs an accounting that the geometry can check. In a dynamical spacetime a flux of energy through a surface is not a conserved mass unless a time-flow vector and a hypersurface are specified. The unambiguous quantity in spherical symmetry is the Misner–Sharp mass of Eq. (3), which is

conserved in vacuum and changes only by the flux of matter, and the exterior mass parameter $m_{ext}=2GM$ that it defines. In the small-tension limit the critical mass $m_c \to (1+q)R_{dS}$ has a simple meaning: it is $\Delta R_{dS}^3/3$, the vacuum-energy difference $\Delta=(1+q)V_f$ integrated over a de Sitter volume of radius R_{dS}, $M_c=(4\pi/3)\Delta R_{dS}^3$. A false-vacuum region of radius R_w has an exterior mass of at least $\Delta R_w^3/3$ because of its vacuum energy alone, so that $m>m_c$ is automatic for $R_w>R_{dS}$ and no matter is needed to supply it. In the configurations analysed the expanding core is vacuum energy and nothing else: none of the parent's matter enters it, and none has to. Matter is transferred in no sense, and the exterior black hole is what is left of the parent's vacuum-energy deficit.

The horizon of the exterior at the critical mass has a property that deserves a precise statement. The exterior has $f(r)=1-m/r+q\,r^2/R_{dS}^2$ (for $\rho_{out}=-q\rho_{in}$ the anti-de Sitter radius satisfies $L^2=R_{dS}^2/q$), and its horizons obey $r-m+q\,r^3/R_{dS}^2=0$. At $m=(1+q)R_{dS}$ one root is exactly

$$r = R_{dS} \qquad \Rightarrow \qquad \frac{A_{dS}}{A_h} = \left(\frac{R_{dS}}{r}\right)^2 = 1 \tag{11}$$

for every $q>-1$: the exterior horizon has the area of the interior de Sitter horizon. For $q\geq 0$ this root is the black-hole horizon. For $-1<q<0$ it is the exterior's cosmological horizon and the black-hole horizon is smaller (for $q=-0.5$ at $m=0.5R_{dS}$ the roots are $0.618R_{dS}$ and R_{dS}), so the coincidence is then with the cosmological horizon. At finite tension the critical mass is slightly smaller (Eq. (9)) and the horizon radius at m_c is $R_{dS}\{1-\hat{\sigma}^2[1/4-1/6(1+q)]/(1+3q)\}$, so that the equality of Eq. (11) holds only in the limit $\hat{\sigma}\to 0$ (Table 2); for the model's tension the departure is 3.1×10^{-61}.

Equal horizon areas correspond to equal Bekenstein–Hawking and Gibbons–Hawking entropies $A/4G$ of the two horizons [24,25]. That is a geometric coincidence at the limiting threshold, and it is stated as one. It is not a derivation of the threshold from the generalized

second law. Such a statement would need an entropy accounting for the global spacetime, a definition of the horizons that bound the two regions, and the entropy of the matter, none of which is attempted here, and the entropy bound [26] is cited only as context. The coincidence suggests an entropic reading of the threshold; it does not show that thermodynamics independently requires it. The entropy of the observed dark-energy horizon is $S_{dS} = 3.3 \times 10^{122}$ in units of k_B.

7. How large must the region be, and where can it come from?

In the benchmark model with q=1 the exterior mass must exceed m_c=2 R_{dS} (to the accuracy of the limit), that is a mass $M_c = 1.1 \times 10^{23}\ M_{odot}$, for a region of radius R_{dS} = 5.3 Gpc. This mass is 7.7 times the matter within one Hubble radius today ($1.4 \times 10^{22}\ M_{odot}$) and 4.4 times the matter within a sphere of radius R_{dS} ($2.5 \times 10^{22}\ M_{odot}$), so that today's matter provides only 23% of it. The present-day matter content of that region is insufficient to reproduce the critical mass through matter alone; the critical gravitational mass is instead dominated by the vacuum-energy difference, which the false vacuum itself supplies, so the criterion does not require matter to be gathered. In the past the matter in such a sphere was larger, and equalled M_c at a=0.61 (Fig. 6b); but then the interior was dominated by matter and not by the false-vacuum energy, which is not the vacuum-dominated expanding core studied here, and with dust inside the threshold radius becomes the Hubble radius of the denser interior, a case that the thin-wall and numerical analyses of this paper did not treat. Table 4 gives the requirement for other depths of the true vacuum; deeper vacua need more.

q	m_c/R_{dS}	M_c (M_{odot})	M_c/M_H	M_c/M_R	Horizon radius r_h/R_{dS} at m_c
0.25	1.25	6.97×10^{22}	4.8	2.77	1.000
0.50	1.50	8.37×10^{22}	5.8	3.32	1.000
1.00	2.00	1.12×10^{23}	7.7	4.43	1.000
2.00	3.00	1.67×10^{23}	11.6	6.65	1.000
4.00	5.00	2.79×10^{23}	19.4	11.08	1.000

Table 4. Critical region for an expanding core, as a function of the depth of the true vacuum ($V_t=-qV_f$), with Planck 2018 parameters and the small-tension limit of the critical mass. M_H is the matter within one Hubble radius today and M_R the matter within a sphere of radius R_{dS}.

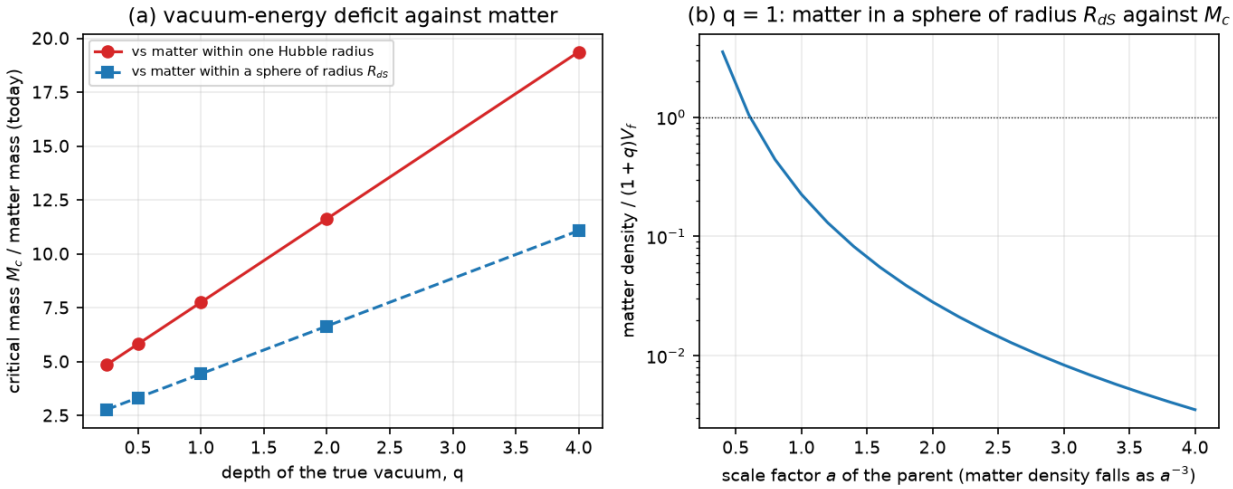


Fig. 6. (a) Critical mass of the false-vacuum region, in units of the matter within one Hubble radius today (red) and within a sphere of radius R_{dS} (blue), against the depth q of the true vacuum. (b) Matter density of the parent, relative to the vacuum-energy difference $(1+q)V_f$ for $q=1$, against its scale factor.

Three conclusions follow, and each can be wrong in a specific way. First, the size: the region has the radius of the de Sitter horizon of the observed dark-energy density, 5.3 Gpc, 1.21 times the Hubble radius today; no lump, void or halo is of that size. A false-vacuum region smaller than $R_{\rm dS}$, whatever its mass, is outcome A: this is what the numerical runs with $R_{\rm w}\lesssim R_{\rm dS}$ show and, for $m<m_{\rm c}$, what the thin-wall analysis proves. Second, the origin: what is left is a region of undecayed false vacuum that is itself larger than $R_{\rm dS}$, which in this model is a patch of the parent universe that has not made the transition. Such a patch is not created by the collapse; the numerical runs show an expanding region only where the initial data already contain it, and what the collapse of its lower-energy surroundings does is trap it behind a horizon. That does not

produce a reset of the parent: it separates a piece of the parent's own vacuum. Third, causality: while the parent accelerates, its event horizon is of order R_{dS}, and a region of radius R_{dS} cannot be assembled by dynamics inside it, because matter beyond the horizon is out of reach; the requirement can be met only by an acausal initial condition or by a patch that survives. The special case in which the whole parent returns to the false vacuum in a contracting phase is not a localized collapse: it is the homogeneous restoration that the model assumes elsewhere, and it needs no throat.

8. Phase diagram and falsification criteria

Collecting the results, the model has three regimes for a false-vacuum region of core radius R_w and exterior mass m, in units of R_{dS} and with q=1:

• $m<m_c$ (thin-wall result): the wall is confined and collapses; a trapped surface forms (outcome A). This includes every region that is smaller than R_{dS} and has no more than its own vacuum energy.

• $m>m_c$ and $R_w \lesssim R_{dS}$: the numerical runs show a trapped surface and a normal core (outcome A), and no expanding region ever appears; extra mass by itself does not keep the core from collapsing.

• $R_w \gtrsim R_{dS}$ (which implies $m>m_c$): the initial data already contain an expanding core, which persists next to a trapped surface (outcome C in the sense of persistence). It requires the horizon-sized region of Section 7, it is not created by the collapse in any run, and it does not transfer matter.

Each statement of the paper can be falsified, and the falsification criteria are:

1. A violation of the null energy condition by the matter that the model contains. This would remove Proposition 1 and reopen outcome B. For the canonical scalar and dust of the model it is excluded; for a model with a non-canonical kinetic term or a ghost it must be checked afresh.

2. A numerical solution, with a thick or thin wall, in which a trapped region and an expanding core coexist at $m_{\mathrm{ext}}<m_{\mathrm{c}}$ with regular initial data. The thin-wall result says that none exists; a counterexample would show that the junction analysis misses a configuration.

3. A run in which an anti-trapped region appears at a later time from initial data that have none. None of the 42 runs with a normal initial core does so; a single instance would turn persistence into creation.

4. A mechanism that forms a vacuum-dominated false region of radius $\gtrsim R_{\mathrm{dS}}$ by the collapse of smaller regions. None is known, and the numerical runs with $R_{\mathrm{w}}\lesssim R_{\mathrm{dS}}$ give none.

5. A thick-wall or global evolution whose threshold differs from the finite-tension $m_{\mathrm{c}}(\sigma)$ by more than the corrections of finite wall thickness. The experiments of Section 5 show the threshold pattern, but do not measure m_{c} to a stated accuracy.

6. A regular, global evolution through the horizon that shows the expanding core is not a separate region, for example by closing the anti-trapped region. The anti-trapped region cannot be undone by the dynamics in the numerical runs, but a global argument has not been given.

9. Discussion

The result is a conditional negative, and the paper has tried to state exactly what is and is not shown. Shown: the null energy condition removes traversable throats from this model; below a finite-tension critical mass every regular configuration collapses; at the limit of vanishing tension the critical mass is $(1+q)R_{\mathrm{dS}}$, at which the horizon areas coincide; trapped surfaces were identified in 65 of the 72 numerical runs at the base resolution and in 6 more after refinement,

514 and the constraints and null expansions converge; and a persistent expanding false-vacuum core
515 coexists with the trapped region when the core is larger than its own de Sitter radius. Not shown:
516 that collapse creates a daughter universe. The expanding core is in the initial data of every run in
517 which it is found, and none of the 42 runs with a smaller core develops one; the simulations
518 show persistence beside a collapse, not creation by it.

519 Four caveats limit even the persistence result. The exterior vacuum of the baseline is anti-de
520 Sitter, which is not the late-time state of a universe that approaches heat death; the experiments
521 with a Minkowski and a lower de Sitter exterior (Section 5.5) support the threshold and the
522 dependence of the expanding core on its size, but are largely unresolved with respect to trapped
523 surfaces, and for a de Sitter exterior of large mass there is no black-hole horizon to hide behind.
524 The dark matter that motivates the transition is not simulated: the experiments evolve the scalar
525 field in a prescribed double-well potential, so the complete matter-triggered process has not been
526 simulated. The expanding core is a cold de Sitter region with the vacuum energy V_{f}. If it were
527 interpreted globally as a daughter universe, producing a hot big bang inside it would need a
528 reheating process, and none is included; and it would not reconnect with the parent: to call the
529 process a cycle one would have to show that the daughter replaces or re-enters the parent, which
530 the geometry does not suggest, because the core is behind the horizon of an exterior black hole
531 whose mass is at least M_{c}.

532 The work also leaves things undone. The numerical solution is spherically symmetric and
533 classical, the walls are not as thin as the physical wall, the evolution ends at the singularity, and
534 the bridge of Section 4 is a thin-wall construction: no spacelike slice through it is built from the
535 numerical solutions. The tunnelling route to a baby universe [10,11] is outside the classical
536 treatment, and a quantum or semiclassical stress-energy that violates the null energy condition

locally could in principle allow a throat; the present model has neither. Perturbations of the exterior and of the wall are not studied, and a nonspherical collapse could form different horizons. A global numerical evolution through the horizon, with excision of the singularity, would settle whether the core is a separate region and would test the bridge directly; it is the most useful extension.

10. Conclusions

The question was whether a matter-dilution-triggered vacuum transition can form a wormhole, a non-singular gravitational bridge to an expanding daughter universe, or whether the energy conditions and causal structure of the model forbid it. For this model the answer has three parts. Ordinary traversable throats are forbidden in this classical canonical-scalar and non-negative-density dust model: the null energy condition holds, and no null ray has a minimum of the areal radius. This does not exclude non-traversable bridges, quantum or semiclassical continuations, or baby-universe scenarios that rest on them. A non-traversable bridge with an expanding region requires an exterior mass above the finite-tension critical value, which is $(1+q)R_{\mathrm{ds}}$ in the limit of a probe wall, where the exterior and interior horizon areas coincide; and in the numerical runs the expanding region is present only where the initial data already contain a false-vacuum core larger than its own de Sitter radius, so that the runs show persistence and not creation. At the observed dark-energy scale that core has a radius of 5.3 Gpc and a vacuum-energy deficit of $1.1 \times 10^{23}\, M_{\mathrm{odot}}$, 4.4 times the matter within a sphere of that radius today. The critical mass is dominated by the vacuum-energy difference and not by matter; a localized collapse cannot supply a region of that size, and the matter of the parent does not pass into the core. The threshold itself is established for the thin-wall junction branches with regular initial data; for thick walls the experiments show its pattern but do not measure m_{c}, the physical wall is

thinner than any numerically resolved wall by thirty orders of magnitude, and for other exterior cosmologies only the thin-wall statement and the incomplete experiments of Section 5.5 apply. The mechanism is therefore best described as the isolation of a horizon-sized false-vacuum patch, not as a cyclic reset, and it is not supported as an end-of-universe process of the model.

Declarations

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Competing interests: the author declares no competing interests.

Author contributions (CRediT): Leon Sandler, sole author: conceptualization, methodology, software, formal analysis, writing, review and editing.

Data and code availability: the code, the data of every figure and table and the manuscript are archived at https://doi.org/10.5281/zenodo.23995982 (software, all versions) and https://doi.org/10.5281/zenodo.23995978 (manuscript, all versions), and are available at https://github.com/sandlerleon/vacuum-transition-throats. One command per script regenerates every number and figure; the results quoted in the text are read from the JSON files that the scripts write.

Use of generative AI: a generative AI tool (Claude, Anthropic) assisted with code development, numerical experiments, drafting and editing. The author checked all of it and takes full responsibility for the content.

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